Problem Set 7: Development, Labor & Education

Microeconomic Theory for Experimentalists & Applied Microeconomists

Due: [**date — before Lecture 1 of Module 8**]

Ground rules: collaboration on ideas is encouraged; write-ups are individual. You may use AI tools for parts flagged with ⟨AI⟩ and must attach the transcript. Show all calculations.

Weights: Part A — 30 points. Part B — 45 points. Part C — 25 points.

Part A: Theory — Guided Calculations (30 points)

Question 1 (Dynamic complementarity with numbers — 15 points)**.** *A child's adult skill depends on early-childhood investment I_E and later investment I_L. Consider two candidate technologies:* complements, $S = I_E \times I_L$, *and* substitutes, $S = I_E + I_L$. *A family (or a government) has 10 units to allocate.*

(a) ***(4 pts)*** *Under each technology, find the allocation of the 10 units that maximizes S, and the resulting skill level (if the maximizer is not unique, say so).* Hint: under complements, think symmetry.

(b) ***(4 pts)*** *Now suppose early investment is stuck at $I_E = 2$ (the early window was missed). Under the complements technology, how many units of late investment are needed to reach the skill level 25 that a balanced 5/5 allocation would have produced? What is the total cost of that remediation path (count both the early and the late units), and what is the marginal product of a late unit at $I_E = 2$ versus at $I_E = 5$?*

(c) ***(4 pts)*** *In one paragraph: why do late-remediation programs systematically underperform early programs* if *the technology is complementary, and what 2×2 experiment identifies whether it actually is? State the estimand in one phrase (Module 6 named it).*

(d) ***(3 pts)*** *A skeptic notes that production-function parameters estimated from observational data are contaminated by* parental responses*: parents may reinforce early advantages or compensate for early deficits. Explain in two sentences how each response biases the observational estimate of complementarity, and why randomization in the 2×2 experiment of (c) removes this bias (parental responses to the programs themselves remain part of the measured effect — Part B(c) returns to this).*

Question 2 (Deworming, externalities, and cost-effectiveness — 15 points)**.** *A deworming program costs \$0.50 per treated child per year. Evaluation evidence (stylized Miguel–Kremer magnitudes) shows each treated child gains 0.075 school-years per year of treatment. In addition, each treated child generates an infection-reduction externality: 2 nearby untreated children each gain 0.02 school-years.*

(a) ***(5 pts)*** *Compute the cost per additional school-year (i) counting only the direct effect and (ii) counting direct plus externality effects. By what factor does ignoring the externality overstate the cost?*

(b) ***(4 pts)*** *The externality was only visible because randomization was at the* school *level. Explain in three sentences why child-level randomization within schools would have (i) contaminated the control group and (ii) biased the estimated treatment effect, stating the direction of the bias.*

(c) ***(3 pts)*** *A ministry ranks programs by cost per school-year and funds the top three. Using your numbers, explain how the choice of randomization level in the* evaluation *can change which programs get* funded *— one concrete sentence about the ranking.*

(d) ***(3 pts)*** *Name one other intervention type (any domain) whose evaluation plausibly zeroed out a spillover by randomizing at too fine a level, and say in one sentence what the coarser design would be.*

Part B: Experimental Design (45 points)

Question 3 (Testing dynamic complementarity in the field)**.** *Design a field experiment in a low-income setting that tests whether early and later child investments are complements — not merely whether each program "works." Specify:*

(a) ***(6 pts)*** *Setting, subject pool, and recruitment: the two investment programs (an early-childhood program, ages 1–3, and a later program, ages 5–7) and why your setting makes the complementarity question policy-relevant.*

(b) ***(12 pts)*** *Treatments: the cross-randomized 2×2 (neither / early only / late only / both), what each comparison identifies, and why the* interaction *is the estimand — including the prediction each technology (complements vs. substitutes) makes for the "both" cell relative to the sum of the single-program effects.*

(c) ***(9 pts)*** *Measurement: child skill assessments at multiple ages AND parental investment responses (time, money, attention) — explain why measuring parental responses is not optional in this design (Part A(1)(d) is the reason), and how you will distinguish reinforcement from compensation in the data.*

(d) ***(9 pts)*** *Hypotheses and power: state the interaction hypothesis under each technology, and — using Module 6's lesson that interactions need roughly 4× the sample of a main effect — give per-cell and total sample sizes anchored to a stated main-effect size (e.g. 0.25 SD per program, typical of early-childhood interventions; any reasoned anchor accepted). Note the follow-up horizon problem and your interim strategy.*

(e) ***(9 pts)*** *Ethics with a vulnerable population — this is graded as designed substance, not boilerplate: equipoise for each cell (why is "neither" ethical here?), what the control group receives, consent under low literacy, local approval and benefit sharing, and one design change you would accept at an ethics board's request even at a power cost.*

Part C: Silicon Pilot Interpretation ⟨AI⟩ (25 points)

Question 4 (The population that matched the mean)**.** *You run `module7-simulation-lab.ipynb`. Your synthetic rural-Kenya population's mean dictator-game giving lands within one percentage point of the published human estimate you entered as the calibration target (you did not tune the population to the target; the match arose on the first run). Your lab partner declares the population "calibrated" and proposes piloting a cash-transfer instrument on it to forecast the transfer's consumption response for the grant proposal.*

(a) ***(8 pts)*** *Using the calibration hierarchy (marginals → behavioral moment → gradients/heterogeneity → treatment responses), explain what the matched mean does and does not establish, and give two LLM-specific mechanisms that can produce a matched mean WITHOUT the population being calibrated in any useful sense.*

(b) ***(9 pts)*** *Design the diagnostic battery for the next two rungs: (i) a gradient check (which demographic gradients, from which human literature, and what pattern confirms vs. refutes calibration) and (ii) an out-of-task check (a second game or elicitation with published cross-cultural data). State what result pattern would justify which uses of the population. Run it if API budget allows and report; otherwise state predictions precisely.*

(c) ***(8 pts)*** *The grant proposal: state one thing the synthetic population CAN legitimately contribute to the cash-transfer study (and why), one thing it CANNOT (and the circularity that makes rung 4 unreachable from inside the sim), and the sentence you would write in the proposal's methods section describing the silicon pilot's role honestly.*

Attach your AI transcripts and, if you ran code, the results CSV.